

Provenance Is Enough: Removing Backdoors From the Parameters an Untrusted Source Shaped

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Abstract—Training corpora are assembled from sources a defender trusts unequally: an audited internal feed, a partner’s contribution, a scrape nobody has vetted. That knowledge is coarse, it is already recorded, and backdoor defences do not use it. We show it is enough. We attribute every parameter of a trained network to the training samples that formed it, weighted by their share of its gradient, and carry the defender’s opinion of each source onto the parameters. Removing the parameters an untrusted source shaped drives a patch backdoor’s attack success from 99.08% to 0.10% for 0.15 points of clean accuracy on MNIST, without retraining, without a clean reference set, and without discarding the source’s benign data. The defence needs no knowledge of the trigger, and whether it applies is decidable before training from the variance of trusted data alone: it holds where a trigger must sit on a region clean data leaves quiet, which MNIST and Fashion-MNIST have and GTSRB does not.

Its strength is the assumption it rests on. At a 0.97% poisoning rate, where activation clustering removes nothing and spectral signatures recover 1% of the poison, provenance is exactly as informative as before and the backdoor is still removed; sample-level influence ranking, the expected alternative, catches less poison than random removal at either rate. We are precise about what a defender’s representation of provenance must preserve. With a single untrusted source every criterion coincides, which we prove. Against an adaptive attacker who also hides poison in the audited source, a score that only separates verified from unverified data fails completely between 30% and 40% contamination, while any score that keeps *unvetted* and *compromised* distinct removes the attack. A subjective-logic opinion is one such score and additionally records how much evidence each parameter carries. Attributing to input positions rather than parameters extends the defence to a ResNet-18 on four datasets, GTSRB included, reproduces the adaptive-adversary result there with a wider window, and we report where random-crop augmentation defeats it.

Index Terms—backdoor defence, data poisoning, data provenance, subjective logic, trust

I. INTRODUCTION

Training data is assembled, not collected. A practitioner combines an audited internal corpus with a partner feed, a scraped supplement and whatever a data vendor supplies, and knows quite a lot about *where* each part came from while knowing almost nothing about what is wrong inside any of it. That knowledge is coarse, it is already recorded, and backdoor defences do not use it. Defences that separate poisoned samples in a learned representation [1], [2] need the corruption to be visible there; defences that prune dormant units [3] need a clean reference set; methods that trace a

model’s behaviour back to its training samples [4], [5] score each sample on its own and never ask where it came from.

We show that the coarse knowledge is enough. A parameter’s gradient is a sum of per-sample contributions, so weighting each contribution by the defender’s opinion of the sample’s source yields an opinion about the parameter. Parameters that only an untrusted source ever moved inherit its status; parameters that audited data shaped do not. Removing the former removes an implanted backdoor without retraining, without a clean reference set, without knowledge of the trigger, and without discarding the untrusted source’s benign data: on poisoned MNIST, attack success falls from 99.08% to 0.10% for 0.15 points of clean accuracy.

Three properties of this defence are, as far as we know, unusual in the literature, and they are what the paper contributes.

The first is that its applicability is decidable before training. The defence removes parameters, so it can only act on corruption that concentrates in a few of them, and a patch trigger concentrates only where clean data leaves the input quiet. Whether such a region exists is a property of the trusted data alone. MNIST and Fashion-MNIST have one and GTSRB does not; the defence works on the first two and correctly declines on the third, and a defender can learn which case they are in before any model exists.

The second is that its strength does not depend on the attack being conspicuous. At a 0.97% poisoning rate the backdoor still succeeds 95.7% of the time, but the poisoned samples no longer form a cluster that activation clustering accepts, and spectral signatures recover 1% of them. Provenance is exactly as informative as before, and the backdoor is still removed. Influence-ranked sample removal, the natural alternative built from the same gradients, catches less of the poison than removing samples at random, because self-influence is largest for atypical samples and poisoned samples are deliberately easy.

The third is that we can say precisely what a defender’s representation of provenance must preserve. With a single untrusted source, every criterion built from the attribution induces the same ranking, which we prove, so how trust is represented is immaterial there. It stops being immaterial against an adaptive adversary who also hides poison in the source the defender believes audited. A parameter carrying the trigger then acquires verified evidence alongside compromised evidence, and a score that only separates verified from unverified

fied data ranks benign unvetted parameters as more suspicious than the attacker’s own: between 30% and 40% contamination it recovers none of the trigger and the backdoor survives at 99%. Any score that keeps *unvetted* and *compromised* data distinct removes the attack across that window. We represent provenance as subjective-logic opinions [6], which keep belief, disbelief and uncertainty on separate axes and make the distinction explicit. A scalar built from the same three states does as well, and we report that rather than claim more for the formalism than it earns; what the opinion adds is a record of how much evidence each parameter carries, which the scalar discards.

Weight sharing detaches a convolutional filter from any image location, so “the parameters an untrusted source shaped” has no spatial meaning there. Attributing to input positions instead, and neutralising flagged positions at inference, carries the defence to a ResNet-18 on four datasets, GTSRB included, and adds a condition of its own: the trigger must sit at fixed coordinates during training, which random-crop augmentation violates.

Contributions:

- A backdoor defence that consumes only source-level provenance, needing no clean reference set, no knowledge of the trigger and no retraining, and that removes the backdoor at a conspicuous and at a quiet poisoning rate where representation-space defences fail at the latter (Sections III and V-A).
- Its applicability condition, measured, and decidable in advance from trusted data alone (Section V-C).
- A statement of what a representation of provenance must preserve: a proposition that every criterion coincides for a single untrusted source, and an adaptive-adversary experiment in which two-state scores fail and every three-state score succeeds (Sections III-D and V-D).
- An extension to convolutional models by input-position attribution, working on all four datasets on a ResNet-18, reproducing the adaptive-adversary result on that architecture with a wider window, and with the augmentation boundary reported (Section VI).
- Boundary reporting throughout: the decision rule’s false alarms and their cost, the trigger-size limit, and every failure case we found.

II. THREAT MODEL

Asset: The integrity of the deployed classifier’s predictions. The defender wants a model that does not carry behaviour an adversary can trigger at will.

Adversary: A training-time poisoning adversary who controls one contribution to the training corpus. Following standard backdoor practice [7], [8], the adversary adds a fixed trigger pattern to a subset of the samples they control and relabels them, so that inputs carrying the trigger are classified as the adversary chooses while accuracy on clean data is unaffected. The adversary chooses the trigger, its size and position, the targeted classes, and the poisoning rate within their contribution. They do not control the architecture,

the training procedure, or the defender’s provenance records. In Section V-D we strengthen this: an *adaptive* adversary additionally compromises a fraction of the source the defender believes is audited, so the provenance records are partly wrong.

Defender knowledge: The defender holds the trained model, the training corpus, and a partition of that corpus into sources with an opinion about each. That opinion is coarse. It says a source was audited, or that its provenance is unknown, or that it is believed compromised. The defender does *not* know which samples within any source are poisoned, what the trigger looks like or where it is, which classes are targeted, or the poisoning rate. They have no clean reference set beyond what provenance already identifies, and no held-out data known to be trigger-free.

Defender capability: The defender may inspect and modify parameters. We treat retraining as available but costly, and report it explicitly for every arm that uses it, because in practice retraining may be impossible: the model may be delivered as an artefact, the original pipeline may be gone, or the compute may be out of reach.

Out of scope: Evasion attacks at inference, model stealing, membership inference, and adversaries who control the training procedure itself. We make no claim of a formal robustness guarantee; the results are empirical, and the conditions under which the defence fails are reported in Section VII.

III. METHOD

A. Opinions over sources

We represent trust as a subjective-logic binomial opinion [6] $\omega = (b, d, u)$ with $b + d + u = 1$, denoting belief, disbelief and uncertainty. The projected probability is $P(\omega) = b + a u$ with base rate $a = 1/2$ throughout. The representation matters here for one reason: $(0, 0, 1)$ and $(0, 1, 0)$ are different statements. The first says nothing is known about a source; the second says it is believed bad. A scalar score collapses them.

The defender supplies one opinion per source. An audited feed is $(1, 0, 0)$, a feed of unknown provenance is $(0, 0, 1)$, a feed believed compromised is $(0, 1, 0)$. Sample n inherits the opinion of its source, written (b_n, d_n) .

B. Attribution: from samples to parameters

For a dense layer l with input activation $a^{(l)}$ and back-propagated error $\delta^{(l)}$, the gradient of parameter $\theta_{ji}^{(l)}$ is a sum of per-sample outer products, $\sum_n a_{j,n}^{(l)} \delta_{i,n}^{(l)}$. The magnitude of each term is how much that sample moved that parameter. We use it as an attribution weight and accumulate, over the whole training set,

$$m_{ji} = \sum_n |a_{j,n}^{(l)}| |\delta_{i,n}^{(l)}| \quad (1)$$

$$r_{ji} = N \sum_n |a_{j,n}^{(l)}| |\delta_{i,n}^{(l)}| b_{j,n} / m_{ji} \quad (2)$$

$$s_{ji} = N \sum_n |a_{j,n}^{(l)}| |\delta_{i,n}^{(l)}| d_{j,n} / m_{ji} \quad (3)$$

where m_{ji} is the total influence mass on the parameter and r_{ji}, s_{ji} are the positive and negative evidence carried by that mass, scaled to a fixed budget N . Mapping the evidence pair

to an opinion by the standard evidential quantification, $\omega_{ji} = (\frac{r}{r+s+W}, \frac{s}{r+s+W}, \frac{W}{r+s+W})$ with non-informative prior weight $W = 2$, gives one opinion per parameter.

Three properties follow directly. A parameter moved only by audited data gets $s = 0$ and high belief. One moved only by compromised data gets $r = 0$ and high disbelief. One moved only by data of unknown provenance gets $r = s = 0$ and remains vacuous: it has no evidence for or against, rather than balanced evidence, and a score that only separates verified from unverified data cannot express that state at all.

For the input layer the opinion is applied per feature, since the defender may hold different opinions about different parts of an input. For hidden layers and bias terms the sample’s own opinion is used, because the provenance of an activation is the provenance of the sample that produced it.

Centring the attribution weight: Equation (1) weights a sample’s contribution by $|a_{j,n}^{(l)}|$, and on standardised inputs that is the wrong quantity. A pixel blank in every clean image does not have magnitude zero after standardisation; it sits at $-\mu_j/\sigma_j$, so clean data appears to drive a parameter that only the poison actually moves, and how badly depends on constants that differ per dataset. We therefore weight the input layer by each sample’s *deviation* from the trusted-data mean, $|a_{j,n}^{(0)} - \mu_j^{\text{tr}}|$, which the defender can compute from the sources they trust. The live filter continues to use the uncentred influence, since a trigger parameter carries little centred mass while being highly influential. This single change is what lets the method transfer between datasets: without it the untrusted share of a trigger parameter’s influence on Fashion-MNIST is 0.344 against a background of 0.342, and with it 0.970 against 0.372.

We compute Equations (1) to (3) as a single replay of the training set through the converged model, which requires no change to the training pipeline and can be run on a delivered artefact.

C. Flagging: distribution-relative, not absolute

Ranking parameters by $P(\omega_{ji})$ and taking a fixed threshold does not work, and the reason is worth stating because it is easy to miss with a single training run. Across three seeds of an identical setup, the median attributed trust of the live features moved from 0.46 to 0.61. A threshold tuned on one model flags everything on another and nothing on a third. What is stable is the *separation*: on all three seeds the trigger features were the lowest-trust live features by a wide margin.

We therefore flag relative to the distribution. The obvious way is a fixed-deviation threshold: writing μ for the median and σ for the scaled median absolute deviation over live features, flag feature j when

$$z_j = \frac{\mu - \min_i P(\omega_{ji})}{\sigma} > k. \quad (4)$$

This works, but k is a distance and distances move with the poisoning rate: Section IV-A measures the dependence and shows the constant that is right at one rate misses the trigger at another. We therefore use a rule with no distance in it. A

corrupted model’s trust values are bimodal, a small cluster the untrusted data dominated and the bulk, so we split the live features where between-class variance is maximal, which is Otsu’s criterion and needs no parameter, and accept the split only when it is genuinely bimodal, that is when between-class variance exceeds twice within-class variance. The one remaining constant is a variance *ratio*, which is scale-free, and a cap of 30% of features guards against a degenerate split. The rule is sensitive: it also fires on models with no backdoor, selecting a comparable number of features. What separates the two cases is which features it selects and what removing them costs, and Table IV reports both.

All results in Sections V, V-A, V-C and V-D use this rule. Section VI uses Equation (4) instead, because on convolutional models the bimodality test fires on only one seed of three; we report that as a limitation of the rule rather than papering over it.

Figure 1(a) shows the distribution the rule acts on. A feature is *live* when its total influence mass exceeds the 20th percentile, which excludes parameters the network never exercises; Section VI-E explains why that filter is dropped in the convolutional setting.

D. What the representation must preserve

Before using the machinery it is worth knowing when it adds nothing, and which part of it does the work when it does.

Proposition 1. *Let every sample come from one of two sources, trusted with opinion $(1, 0, 0)$ or untrusted with a fixed opinion (b_u, d_u, u_u) . Let $f_{ji} \in [0, 1]$ be the untrusted share of parameter θ_{ji} ’s influence mass. Then $P(\omega_{ji})$ is a strictly monotone function of f_{ji} alone, so ranking parameters by attributed trust and ranking them by untrusted influence share produce the same order.*

Proof. From Equations (2) and (3), $r_{ji} = N((1-f_{ji})+f_{ji}b_u)$ and $s_{ji} = Nf_{ji}d_u$, since trusted samples contribute belief 1 and disbelief 0. Both are affine in f_{ji} , hence so are $r+s$ and, through $P = (r+W/2)/(r+s+W)$, the projected probability, whose derivative in f_{ji} has constant sign determined by $b_u + d_u/2 - 1 < 0$. A monotone transform preserves order. $\square$

The same argument covers any source model in which belief and disbelief sum to a constant, including per-feature conformity opinions. With one untrusted source, then, the opinion, the untrusted share and any monotone function of it are the same criterion, and a defender in that setting should use the simplest.

Representations can only diverge when sources carry different *amounts* of evidence, and then what matters is how many states a per-sample score keeps. Take three sources: audited, $(1, 0, 0)$; unvetted, $(0, 0, 1)$; and believed compromised, $(0, 1, 0)$. A score that reads a parameter’s *verified share* of influence, R/m , sees the last two as one state, “not verified”. A

score that reads the influence-weighted mean of each sample’s projected probability,

$$\bar{P}_{ji} = \frac{\sum_n w_{ji,n} P(\omega_n)}{m_{ji}} = \frac{1}{2} + \frac{1}{2} \frac{R_{ji} - S_{ji}}{m_{ji}}, \quad (5)$$

where $w_{ji,n} = |a_{j,n}| |\delta_{i,n}|$ and R, S are the unnormalised sums in Equations (2) and (3), assigns the three sources 1, $\frac{1}{2}$ and 0 and keeps them apart with a single number. The opinion keeps them apart too, and differs from Equation (5) in two respects only. It aggregates a *ratio* of evidence, $r/(r + s + W)$, rather than a difference, and it carries u , so a parameter with balanced evidence and one with no evidence, both at $P = \frac{1}{2}$, remain distinguishable. Section V-D tests all three against an adaptive adversary. We state the outcome here so that the reader knows what the experiment is for: the two-state score fails, and the two three-state scores coincide. The claim the experiment supports is about what a representation of provenance must preserve, not about opinions over scalars.

E. Removal

Flagged input features have their outgoing first-layer weights set to zero. Nothing is retrained and no data is discarded. When the rule flags nothing, the defence does nothing, which is the correct behaviour when the corruption is not localised in input space.

IV. EXPERIMENTAL SETUP

Data and models: MNIST [9] with a 784-128-10 fully connected ReLU network, trained for 20 epochs with batch size 128. Features are standardised with training statistics. All results are over three training seeds unless stated; we report mean and standard deviation, and every arm of every comparison uses the same seeds.

Attack: The trigger is a $p \times p$ patch at the top-left corner set to the maximum feature value, with the labels of the two targeted classes swapped, following [8]. The poisoned contribution is the last third of the training set; within it, only samples of the targeted classes are poisoned, giving 3876 poisoned samples, 6.5% of the corpus. The default trigger is 4×4 , and Section V-A varies it.

Metrics: Clean accuracy on the unmodified test set, and attack success rate (ASR), the fraction of triggered test inputs of a targeted class classified as the adversary’s target, averaged over both directions of the swap. For localisation we additionally report how many of the p^2 trigger features the defence flags, and the AUROC of separating trigger from non-trigger features by attributed trust.

The defender’s opinion about the untrusted source: Unless stated otherwise the untrusted source is given the vacuous opinion $(0, 0, 1)$, *unvetted*, which is what the threat model warrants when a defender knows only that a contribution was not audited. This is not a free choice and we report its effect: asserting $(0, 1, 0)$, *believed compromised*, instead gives every parameter the source touched positive disbelief, which makes the trust distribution bimodal by construction and quadruples the flag set, from 18 to 19 features to 55 to

93, at twice the accuracy cost and with no change in attack success. Section V-D is the exception, where the whole point is that different sources carry different opinions.

Baselines: Every baseline receives exactly the defender knowledge of Section II: which source is untrusted, and nothing more.

- **drop-source:** discard the untrusted source and retrain. The obvious move, and the strongest simple defence available under this knowledge. It costs the benign data that source also carried and a full training run.
- **influence:** rank training samples by self-influence at the converged model, the diagonal term of TracIn [5], drop the most influential, retrain. For a dense layer the per-sample gradient is an outer product, so its squared norm is $\|a^{(l)}\|^2 \|\delta^{(l)}\|^2$ and the score is exact without materialising per-sample gradients. This is the method a reader will reasonably expect us to be reinventing.
- **fine-pruning** [3]: prune the hidden units with the lowest mean activation, given the trusted subset as the clean reference set that the method requires.
- **spectral signatures** [1]: within each class, project the centred penultimate representations onto their top singular vector, drop the largest projections, retrain.
- **activation clustering** [2]: within each class, reduce the representations and split them in two; treat the smaller cluster as poisoned when the split is uneven, drop it, retrain. The method chooses its own removal size, so it is reported with the count it selects. It uses no provenance at all, so it receives less knowledge than we assume and is a demanding comparison.
- **random:** drop the same number of samples, or zero the same number of features, at random. The control that separates a real ranking from the effect of simply training on less data.

Poisoning rates: We report two. The default poisons every eligible sample in the untrusted source, giving 3876 poisoned samples or 6.5% of the corpus. The second poisons 15% of those, giving 581 samples or 0.97%, a quieter attacker that still achieves 95.7% attack success. The two rates separate the defences sharply.

Implementation: Attribution and removal are implemented in NumPy over the trained weights and run on CPU. Training uses PyTorch. Code and all experiment scripts accompany the submission.

A. Why the flagging constant was abandoned

Equation (4) has one free constant, and this subsection is the measurement that made us replace it. It can be calibrated without attack knowledge, since a model trained on the trusted sources alone is by definition backdoor-free and a correctly calibrated k must flag nothing on it. Table I sweeps k over three seeds of both a backdoored and a clean model.

At $k = 8$ the defence flags exactly the sixteen trigger features on every backdoored seed and nothing at all on every clean one: perfect precision and recall on both sides.

TABLE I
CALIBRATION OF THE FLAGGING CONSTANT OVER THREE SEEDS.
ENTRIES ARE FEATURES FLAGGED, WITH TRIGGER FEATURES FOUND IN
PARENTHESES. $k = 8$ FLAGS EXACTLY THE TRIGGER ON EVERY
BACKDOORED MODEL AND NOTHING ON ANY CLEAN ONE.

k	Backdoored (seeds 0/1/2)			Clean (seeds 0/1/2)		
5	17(16)	16(16)	21(16)	14	9	4
8	16(16)	16(16)	16(16)	0	0	0
12	16(16)	0(0)	16(16)	0	0	0
20	0(0)	0(0)	0(0)	0	0	0

But the constant is not universal, and the dependence is systematic. A quieter attacker moves the trigger parameters less, so the separation shrinks and a larger k misses it: at a 0.97% poisoning rate $k = 8$ finds the trigger on only one seed of three, while $k = 4$ finds all of it on all three. A defender does not know the poisoning rate, so this is not a constant they can set. What does not change is the ranking: at both rates the trigger features are the lowest-trust live features, occupying ranks 0 to 18 on every seed.

That is the argument for a rule whose only constant is a variance ratio rather than a distance, and it is why every result we report uses the bimodality split of Section III rather than Equation (4). We keep this table because the fixed-deviation rule is the obvious thing to try and because its failure mode, a constant that has to track a quantity the defender cannot observe, is the concrete reason to prefer a parameter-free test.

V. RESULTS

A. Removal against what a defender would otherwise do

Table II is the main result. Every arm receives the defender knowledge of Section II; the representation-space defences need even less, since they use no provenance at all. We report two poisoning rates, because they separate the methods completely.

At the higher rate, zeroing the eighteen or nineteen input features of 784 that the rule selects drives attack success from 99.08% to 0.10% for 0.15 points of clean accuracy, without retraining. We do not claim this is the best number in the table. Activation clustering reaches 0.03% with slightly better clean accuracy, and discarding the untrusted source reaches 0.02%. Both retrain, and activation clustering additionally needs no provenance at all, so at this poisoning rate it is the stronger defence and we say so.

Two arms fail outright at both rates. Influence-ranked sample removal catches 3.5% of the poison at a 15% budget, less than the 15.5% that random removal catches by construction, and the backdoor survives while clean accuracy falls two points. Self-influence is largest for hard and atypical samples, whereas poisoned samples are deliberately easy ones the network fits comfortably, so the ranking looks in the wrong place. This matters because sample-level influence is the natural thing to reach for; the same gradient information aggregated at *parameter* level localises the trigger exactly. Spectral signatures find much of the poison at the higher rate,

84.8% at a 15% budget, but not enough of it: the residue is sufficient to keep the backdoor at 92.57%.

Low poisoning rates separate the methods: The right-hand block of Table II poisons 0.97% of the corpus. The backdoor still works, at 95.73%. Activation clustering now removes *nothing*: at this rate the poisoned samples no longer form a cluster its criterion accepts, so the defence declines to act and the attack is untouched. Spectral signatures recover 1.1% of the poison. Our defence is unaffected in kind: it flags 18 to 20 features, catches all of the trigger, and drives attack success to 0.03% for the same 0.15 points of clean accuracy it costs at the higher rate. Provenance does not become less informative as the attacker becomes quieter, whereas a representation-space signature does.

This is the sharpest statement we can make about where the method belongs. It is not the strongest defence against a conspicuous attack. It is the only one here that neither retrains nor depends on the corruption being visible in a learned representation, and those two properties are exactly what a quiet adversary and an inaccessible training pipeline take away from the alternatives.

Trigger size: Table III varies the trigger. Small triggers are removed almost for free. A 10×10 trigger spans a hundred features, needs a budget large enough to cover them, and cannot be removed without real cost.

Fine-pruning does not remove this backdoor at any budget we tried. The trigger weights feed hidden units that also serve legitimate features, so those units are not dormant on clean data and activation-based pruning does not reach them.

B. Localisation

Table IV reports what the defence flags across conditions, and what those flags cost. Every row uses the same rule and the same provenance as the removal experiment, so the two tables are comparable.

The rule is sensitive rather than silent, and the cost column is what makes that acceptable. It selects around twenty of 784 features in most conditions, including on models with no backdoor at all, but those features carry so little influence that removing every one of them costs at most 0.13 points of clean accuracy and on one clean model improves it. What distinguishes the backdoored case is not that the rule fires, it is *what* it selects: an AUROC of 1.000, with the sixteen trigger features occupying the bottom of the ranking.

Two rows deserve comment. The control that uses no provenance at all flags nothing and sits at chance, confirming that the signal comes from the defender’s knowledge rather than from the attribution machinery producing suspicious-looking numbers on its own. The trigger-only row carries the patch but correct labels, so the network never learns to exploit it, and the defence flags those features anyway. That is the behaviour the construction implies rather than a false alarm: those positions were shaped by the untrusted source and by nothing else, which is exactly what the attribution measures. A defender who wants a defence that acts only on *exploited* corruption is asking a different question than the one

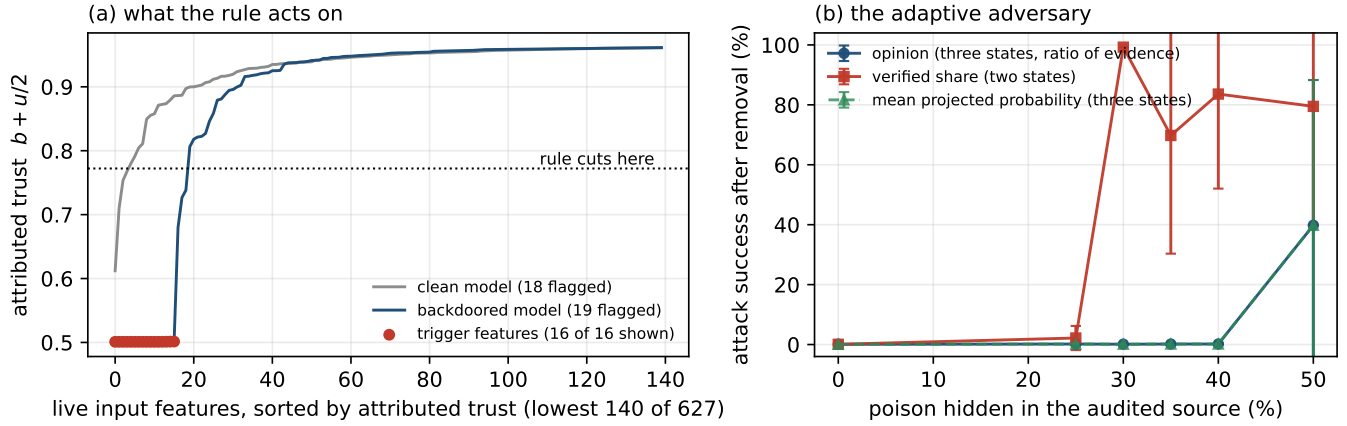


Fig. 1. (a) Attributed trust of the input features of a backdoored MNIST model, sorted, lowest 140 of 627 live features shown, against a clean model from the same pipeline. The sixteen features the trigger feeds form a flat cluster at pure vacuity, far below the bulk; the clean model has no such cluster. The dotted line is where the bimodality rule cuts, and the flag counts in the legend are those of Table IV. (b) Attack success as an adaptive adversary moves poison into the source the defender believes audited, five seeds per level. A two-state score, the verified share of influence, fails between 30% and 40% contamination; the opinion and a three-state scalar, the mean projected probability of Equation (5), remove the attack there and coincide at every level. Beyond 40% none is reliable.

TABLE II

BACKDOOR REMOVAL AT TWO POISONING RATES, THREE SEEDS, MEAN $\pm$ STD. OURS IS THE ONLY ARM THAT NEITHER RETRAINS NOR DISCARDS DATA. AT 6.5% POISONING THE REPRESENTATION-SPACE DEFENCE IS EXCELLENT; AT 1% IT REMOVES NOTHING AT ALL, WHILE OURS STILL REMOVES THE BACKDOOR.

Defence	Retrain	Poisoning rate 6.5%			Poisoning rate 0.97%		
		Clean acc. (%)	ASR (%)	Poison caught	Clean acc. (%)	ASR (%)	Poison caught
None	—	97.68 $\pm$.05	99.08 $\pm$.11	—	97.64 $\pm$.16	95.73 $\pm$.18	—
Ours	no	97.52 $\pm$.05	0.10$\pm$.04	100%	97.49 $\pm$.13	0.03$\pm$.02	100%
drop-source	yes	97.39 $\pm$.12	0.02 $\pm$.02	100%	97.37 $\pm$.11	0.03 $\pm$.02	100%
activation clustering [2]	yes	97.65 $\pm$.04	0.03 $\pm$.05	100%	97.64 $\pm$.16	95.73 $\pm$.18	0%
spectral signatures [1]	yes	97.32 $\pm$.07	92.57 $\pm$ 5.4	84.8%	96.85 $\pm$.32	96.89 $\pm$.26	1.1%
influence [5]	yes	95.67 $\pm$.03	98.79 $\pm$.13	3.5%	95.79 $\pm$.05	94.92 $\pm$.43	14.3%
random	yes	97.46 $\pm$.02	99.17 $\pm$.10	15.5%	97.57 $\pm$.10	96.67 $\pm$.25	14.9%

TABLE III

EFFECT OF TRIGGER SIZE. BUDGET IS THE FRACTION OF INPUT FEATURES REMOVED; FINE-PRUNING IS GIVEN THE TRUSTED SUBSET AS ITS CLEAN REFERENCE.

Trigger	Defence	Budget	Clean acc. (%)	ASR (%)
1×1	none	—	97.67	99.03
	ours	1%	97.75	0.05
4×4	none	—	97.69	98.98
	ours	2%	97.70	0.10
	fine-pruning [3]	10%	97.52	96.93
10×10	none	—	97.82	99.09
	ours	10%	97.47	3.85
	ours	20%	85.90	0.15

TABLE IV

LOCALISATION ACROSS CONDITIONS, THREE SEEDS. PROVENANCE SUPPLIES ONLY THE IDENTITY OF THE UNTRUSTED SOURCE. AUROC SEPARATES TRIGGER FROM NON-TRIGGER FEATURES BY ATTRIBUTED TRUST; IT IS UNDEFINED WHERE NO TRIGGER EXISTS. COST IS THE CLEAN ACCURACY GIVEN UP BY ZEROING EVERY FLAGGED FEATURE, IN POINTS; A NEGATIVE ENTRY MEANS ACCURACY IMPROVED.

Condition	Flagged	Trigger	AUROC	Cost
Backdoored	18.7	16/16	1.000	0.10
Clean, 784-128-10	18.0	—	—	−0.02
Clean, 784-512-10	19.3	—	—	0.02
Label flip only, no trigger	20.7	—	—	0.13
Trigger only, labels correct	20.3	16/16	1.000	0.07
Label-noise trained	0.0	—	—	0.00
No provenance used (control)	0.0	0/16	0.500	0.00

provenance can answer, and the price of answering ours is the 0.07 points in that row.

C. Where the defence works, and where it does not

The defence transfers to Fashion-MNIST and fails on GT-SRB, and the reason is a property of the data the defender can check before deploying it.

MNIST and Fashion-MNIST both have a corner that clean images essentially never vary: the mean per-pixel deviation there is a thousandth and a hundredth of a typical position. A trigger placed on such a region is moved almost only by poisoned samples, its untrusted influence share reaches 1.000

TABLE V

THE APPLICABILITY CONDITION. UNTRUSTED SHARE IS THE FRACTION OF A PARAMETER’S INFLUENCE CONTRIBUTED BY THE UNTRUSTED SOURCE, TRIGGER FEATURES AGAINST TYPICAL LIVE FEATURES. QUIET REGION IS THE MEAN PER-PIXEL DEVIATION OVER THE TRIGGER POSITIONS RELATIVE TO A TYPICAL POSITION, COMPUTED ON TRUSTED DATA BEFORE ANY MODEL IS TRAINED.

Dataset	Quiet region	Untrusted share	Trigger ranks
MNIST	$0.001 \times$	1.000 vs 0.519	0–15
Fashion-MNIST	$0.01 \times$	0.970 vs 0.372	0–17
GTSRB	none	0.359 vs 0.356	33–183

and 0.970 against backgrounds of 0.519 and 0.372, and it sorts to the bottom of the ranking. GTSRB has no such region anywhere: traffic-sign images carry content and background variation across the whole frame, the untrusted share of the trigger parameters is 0.359 against 0.356, and the defence correctly declines to flag anything while the backdoor remains at 77.7%.

The condition is therefore about *where the trigger sits relative to the clean data’s variance*, and it is decidable in advance: the defender computes per-feature deviation on the sources they trust, before training, and learns whether any region exists that this defence could protect. That is an unusual property for a backdoor defence, and it is the honest way to state a limitation an attacker can otherwise exploit, since an adversary who places the trigger over high-variance regions removes the signal the method depends on.

The condition binds the parameter-level form specifically. Section VI attributes to input positions instead, and on that GTSRB backdoor the same provenance yields an untrusted share of 0.474 against a typical 0.343 and removes the attack. We report the comparison in Section VI-D and are careful there about what it does and does not establish, since the two pipelines differ in more than the unit of attribution.

D. The adaptive adversary

Proposition 1 says the representation is immaterial for a single untrusted source, and it is: attribution and a scalar untrusted-share criterion produce identical selections at every budget, to every digit. This subsection constructs the case it does not cover, and uses it to measure what a representation of provenance has to preserve.

The defender now holds three sources: one audited, $(1, 0, 0)$; one of unknown provenance, $(0, 0, 1)$, which is benign but is the only provider of some classes; and one believed compromised, $(0, 1, 0)$, which carries the trigger. The adversary is adaptive: besides the source they are known to control, they also place a fraction of the poison in the source the defender believes audited, so the provenance records are partly wrong. Table VI sweeps that fraction for three criteria, each given the budget the opinion’s own rule selects so that the comparison stays matched: the opinion, the two-state verified-share score R/m , and the three-state scalar of Equation (5).

Table VI first answers the question the threat model raises: why would a defender keep data from a source they be-

TABLE VI

ATTACK SUCCESS (%) AS AN ADAPTIVE ADVERSARY MOVES POISON INTO THE SOURCE THE DEFENDER BELIEVES AUDITED. FIVE SEEDS PER LEVEL EXCEPT THE FIRST. TRIGGER RECALL IN PARENTHESES. ALL CRITERIA USE THE SAME RULE AND BUDGET.

Leak	Undef.	two states	three states	
		Verified share	Opinion	Equation (5)
0%	99.00	0.09 (100)	0.09 (100)	0.07 (100)
25%	99.27	2.18 ± 4.0 (95)	$0.17 \pm .05$ (100)	$0.17 \pm .05$ (100)
30%	99.31	$99.25 \pm .07$ (0)	0.11 $\pm .08$ (100)	0.11 $\pm .08$ (100)
35%	99.29	69.78 ± 39.5 (31)	0.16 $\pm .09$ (100)	0.16 $\pm .09$ (100)
40%	99.40	83.60 ± 31.6 (14)	0.17 $\pm .09$ (100)	0.16 $\pm .09$ (100)
50%	99.28	79.49 ± 39.6 (19)	39.80 ± 48.5 (60)	39.82 ± 48.5 (60)

TABLE VII

ATTACK SUCCESS (%) FOR THE DISCARD-AND-RETRAIN BASELINES. NEITHER REMOVES POISON THE ATTACKER PLACED IN THE SOURCE THE DEFENDER TRUSTS.

Defence	0% leak	25% leak	50% leak
drop compromised source	0.02	96.85	98.59
drop all unverified	0.18	97.36	98.83

lieve compromised, rather than drop it? Without leakage they should, and Table VII shows discarding the compromised source removes the backdoor completely at 0.02%. That stops being true as soon as the attacker anticipates it. At a quarter leakage, discarding the compromised source leaves the backdoor at 96.85% and discarding everything unverified leaves it at 97.36% while costing nineteen points of clean accuracy, because in both cases the poison in the audited source survives untouched.

Two states fail, three states suffice: The criteria then separate, and they separate along one line. Up to a quarter leakage all three agree, which is what Proposition 1 leads one to expect while the evidence a parameter carries stays close to homogeneous. Between 30% and 40% the two-state score collapses, consistently over five seeds: it recovers none to a third of the trigger and leaves the backdoor at 69% to 99%. Both three-state scores remove the attack every time, at 0.11% to 0.17% with complete trigger recall, and they do so with the same numbers to the second decimal at every level, including their common failure at half leakage. The opinion and Equation (5) are, on this adversary, the same defence.

The mechanism is worth stating. A two-state score reads a parameter’s verified share. A parameter driven only by the unvetted source has a verified share of zero and looks maximally suspicious, although it is benign. A trigger parameter, once the attacker has leaked poison into the audited source, acquires genuine verified influence and its share rises. Past some contamination the two-state score therefore ranks benign unvetted parameters *below* the attacker’s own and spends its budget on them. A three-state score does not, because the unvetted source contributes neither belief nor disbelief: those parameters sit at $\frac{1}{2}$ rather than at the bottom, and the trigger, carrying real disbelief, still sorts lowest. Absent evidence and negative evidence are different, and between 30% and 40%

contamination that difference is the whole defence.

What this does and does not say about the formalism:

We represent provenance as opinions because they make the three states explicit and because they carry uncertainty, but Table VI is not evidence that the formalism is necessary. A scalar that assigns the three sources 1, $\frac{1}{2}$ and 0 and aggregates by influence does everything the opinion does here. What the opinion retains and the scalar discards is u : two parameters at $P = \frac{1}{2}$, one with balanced evidence and one with none, are the same to Equation (5) and different to the opinion. No experiment in this paper turns on that distinction, and we say so rather than let the machinery take credit for a result that belongs to the representation’s state count.

An honest boundary on both sides: The window is bounded below as well as above. Below 30% leakage a verified-share score does the job and no representation earns anything, which Proposition 1 predicts and we confirm; at 50% no criterion works reliably at parameter level, although the input-position form of Section VI-G still does. We also found the window’s location sensitive to the decision rule rather than to the attribution: an earlier fixed-deviation threshold placed the separation at 25%, and the bimodality rule moved it to 30% by helping the two-state score as much as the others. The ordering of parameters is what the representation controls; how much of that survives into a decision depends on a rule we do not consider solved (Section VI).

VI. CONVOLUTIONAL MODELS

A. Why the unit of attribution has to change

For a dense layer, a parameter belongs to exactly one input feature, so an opinion about a parameter is an opinion about a feature and removal is well defined. Weight sharing breaks that correspondence. A convolutional filter is applied at every position and tied to none of them, so a filter that a trigger taught cannot be attributed to the trigger’s location, and pruning it removes its behaviour everywhere at once.

We therefore attribute to input *positions*, using the gradient of the loss with respect to the input:

$$m_p = \sum_n |x_{n,p} - \mu_p^{\text{tr}}| \left| \frac{\partial \mathcal{L}_n}{\partial x_{n,p}} \right|, \quad (6)$$

with r_p and s_p formed from the same products weighted by the belief and disbelief of each sample, and mapped to an opinion as before. For a dense first layer $\partial \mathcal{L} / \partial x_j$ is exactly the parameter-weighted sum of the errors that feature j ’s parameters receive, so Equation (6) generalises the dense construction rather than replacing it, and it is defined for any architecture. The output is a spatial trust map.

The defence changes with the unit. Parameters cannot be removed per position, so flagged positions are instead neutralised at inference by substituting the trusted-data mean. The model is untouched and nothing is retrained.

B. Setup

The classifier is a ResNet-18 [10] trained from scratch, for 30 epochs on MNIST, Fashion-MNIST and GTSRB and 40

on CIFAR-10, batch size 256, one-cycle schedule with peak learning rate 0.1. The threat model is unchanged: the last third of the training corpus is the untrusted contribution, and within it the samples of the two targeted classes carry a 4×4 corner trigger and swapped labels. CIFAR-10 is RGB, so the same patch occupies 48 input positions rather than 16. We add GTSRB here because it is the dataset the parameter-level defence cannot handle (Section V-C).

We run every dataset twice, with and without random-crop and horizontal-flip augmentation. This is not an ablation of our method: it is the standard recipe for training a residual network to a competitive accuracy, and Section VI-F shows it is where the method’s one architectural condition bites.

C. Results under standard training

Under standard training the defence removes the backdoor on all four datasets, taking attack success to at most 0.13% against floors of the same order, with clean accuracy costing between nothing and half a point.

CIFAR-10 shows the method does not need a wide margin. A ResNet-18 at 87.58% carrying a backdoor at 98.93% is a model worth defending, and all 48 trigger positions are recovered on all three seeds even though the separation is far narrower than MNIST’s: 0.455 against a typical 0.311, rather than 0.999 against 0.328. Attack success goes to 0.13% against a floor of 0.32% for 0.48 points of clean accuracy and no retraining, so colour and depth are not what limits this method.

D. GTSRB: the dataset the parameter-level form cannot do

GTSRB is where Section V-C draws the boundary of the dense defence. Traffic-sign frames carry content and variation everywhere, no region is quiet, the untrusted share of the trigger parameters is 0.359 against a typical 0.356, and the rule correctly declines to act while the backdoor survives at 77.7%.

Attributing to input positions, the same trigger on the same data separates. The untrusted share of the trigger positions is 0.474 against a typical 0.343, 12.3 of 16 positions are recovered, and masking takes attack success from 95.82% to 0.03% against a floor of 0.17% while clean accuracy is unchanged at 97.97%. The clean-model control matters more here than anywhere else, because a rule that simply favoured corner positions would produce this result for the wrong reason: on GTSRB models trained with the same provenance split and no trigger, the rule flags *no* position on any seed, and the untrusted share at the trigger positions is flat at 0.336 against 0.333. The separation exists only when there is something to find.

We are deliberate about what this does not show. The convolutional pipeline differs from the dense one in three ways at once: the unit of attribution (input positions rather than parameters), the architecture (a ResNet-18 at 97.96% rather than a fully connected network), and the intervention (masking with the trusted mean rather than pruning parameters). One comparison cannot say which of the three is responsible, and

TABLE VIII

RESNET-18, THREE SEEDS, INPUT-POSITION ATTRIBUTION. *Standard* IS THE PLAIN TRAINING PIPELINE; *augmented* ADDS RANDOM CROP AND HORIZONTAL FLIP, THE RECIPE A RESIDUAL NETWORK NORMALLY NEEDS. THE FLOOR IS HOW OFTEN A *clean* INPUT OF A TARGETED CLASS IS ALREADY CLASSIFIED AS THE OTHER, WHICH MASKING CANNOT GO BELOW. SHARE IS THE FRACTION OF A POSITION’S INFLUENCE CONTRIBUTED BY THE UNTRUSTED SOURCE, TRIGGER POSITIONS AGAINST TYPICAL ONES. CIFAR-10 IS RGB, SO THE 4×4 TRIGGER OCCUPIES 48 INPUT POSITIONS RATHER THAN 16.

Dataset	Training	Undefended			Found	After masking		Share
		clean	ASR	floor		ASR	clean	
MNIST	standard	99.59	99.98	0.00	16.0/16	0.00	99.59	.999/.328
Fashion-MNIST	standard	93.89	99.93	0.02	16.0/16	0.02	93.89	.963/.298
GTSRB	standard	97.96	95.82	0.17	12.3/16	0.03	97.97	.474/.343
CIFAR-10	standard	87.58	98.93	0.32	48.0/48	0.13	87.10	.455/.311
MNIST	augmented	99.24	100.0	0.00	15.3/16	0.03	99.24	.997/.250
Fashion-MNIST	augmented	94.75	100.0	0.00	14.3/16	33.2	94.74	.760/.232
GTSRB	augmented	95.41	94.08	0.80	0.3/16	93.86	95.41	.516/.318
CIFAR-10	augmented	93.90	99.52	0.43	2.7/48	99.57	92.96	.219/.256

TABLE IX

EFFECT OF THE LIVENESS FILTER, ON THE THREE DATASETS WE RAN UNDER BOTH SETTINGS. DISCARDING THE LEAST INFLUENTIAL FIFTH OF POSITIONS BEFORE SCORING COSTS RECALL IN FIVE OF THE SIX, AND RECALL IS WHAT THE DEFENCE CONVERTS INTO PROTECTION.

	Bottom fifth discarded		All positions scored	
	found	ASR	found	ASR
MNIST	16/16	0.00	16.0/16	0.00
Fashion	5.0/16	84.1	16.0/16	0.02
CIFAR-10	43.3/48	0.12	48.0/48	0.13
MNIST, aug.	6.0/16	99.98	15.3/16	0.03
Fashion, aug.	0.3/16	99.98	14.3/16	33.2
CIFAR-10, aug.	0/48	99.75	2.7/48	99.57

we do not claim it is the unit. What it does establish is narrower and still useful: the placement condition is a limitation of the parameter-level construction and not of provenance-based attribution as such, so a dataset that fails the condition in Table V is not thereby out of reach.

E. Coverage, not ranking, was the binding constraint

The dense pipeline scores only parameters carrying appreciable influence, discarding the bottom fifth as inactive, and we inherited that filter here. On dense models it costs nothing, because recall there is already complete: the trigger features occupy ranks 0 to 15 on MNIST and 0 to 17 on Fashion-MNIST (Table V). On convolutional models it is the wrong default, and Table IX shows the way it fails is instructive.

On Fashion-MNIST with the filter in place, five of sixteen trigger positions were recovered and the backdoor survived at 84.1%. The attribution was not wrong about the other eleven. Their untrusted influence share is 0.963, they simply never entered the ranking, because a trigger position is quiet by construction: it sits where clean data does not vary, which is exactly what makes it look inactive to a filter that measures influence. The filter was removing the positions the method exists to find. Scoring every position recovers all sixteen and takes attack success to 0.02%, and the same change takes augmented MNIST from 6/16 and a live backdoor to 15.3/16 and 0.03%.

Two things make this safe rather than merely permissive. Positions carrying no evidence receive a vacuous opinion, not a distrusted one, so widening the candidate set adds abstentions rather than accusations. And the false alarms it does add are cheap. On a control model trained with the same provenance split and no trigger at all, the rule flags 37.3 of 784 positions on MNIST, 3.0 of 784 on Fashion-MNIST, and none at all on GTSRB or CIFAR-10, and masking every one of them costs at most 0.01 points of clean accuracy. The untrusted share it measures on those models is flat, 0.335 against 0.336 on MNIST and 0.334 against 0.334 on CIFAR-10, which is what a model with nothing to find should look like.

Precision was never the problem. In every setting where the ranking works the trigger positions occupy the very bottom of it, ranks 0 to 15 on Fashion-MNIST and 0 to 89 of 3072 on CIFAR-10. What we had was a recall ceiling imposed by a preprocessing step, and partial recall is not partial defence: masking six positions of a sixteen-position trigger changed attack success by 0.02 points.

F. Augmentation is the remaining boundary

Random crop and horizontal flip, applied for accuracy and not as a defence, are the one condition the convolutional setting adds, because a position-indexed attribution assumes the corruption sits at fixed coordinates and cropping moves it. The effect is not uniform across datasets, and the two failures are not the same failure.

MNIST is unaffected: 15.3/16 positions and attack success 0.03%. Its corners are blank for clean data, so influence there stays attributable to the untrusted source even after translation. Fashion-MNIST works on two seeds of three, which is the 33.2 ± 47.0 in Table VIII and is a way of writing “one seed at 99.7% and two at 0.00%” that should not be read as a mean.

On CIFAR-10 the attribution itself breaks. The untrusted share of the trigger positions *inverts*, 0.219 against a typical 0.256, so the signal does not weaken, it points the wrong way, and the trigger positions rank in the upper half of the trust ordering, between 1121 and 3055 of 3072 on two seeds of three. CIFAR-10 images carry content in every corner, so

TABLE X

THE ADAPTIVE ADVERSARY AT INPUT-POSITION LEVEL, MNIST, RESNET-18, FIVE SEEDS. ATTACK SUCCESS (%) AFTER MASKING, TRIGGER RECALL IN PARENTHESES; UNDEFENDED ATTACK SUCCESS IS 99.93% TO 99.99% AT EVERY LEVEL. ALL CRITERIA USE THE SAME RULE AND BUDGET. *Drop* DISCARDS THE COMPROMISED SOURCE AND RETRAINS.

Leak	two states		three states		Drop
	Verified share		Opinion	Equation (5)	
0%	0.00 (100)		0.00 (100)	0.00 (100)	0.00
25%	2.18±3.0 (68)		0.00 (92)	0.00 (94)	99.86
30%	1.71±1.7 (61)		0.00 (99)	0.00 (99)	99.83
35%	41.2±47 (45)		0.04 ±.08 (94)	0.04 ±.08 (95)	99.83
40%	46.9±33 (36)		0.12 ±.10 (86)	0.07 ±.09 (88)	99.92
50%	60.8±48 (20)		0.01 ±.02 (82)	0.01 ±.02 (82)	99.92

once cropping translates the trigger no fixed coordinate holds a disproportionate share of the untrusted influence, and the backdoor the model learns is location-robust.

GTSRB fails differently. The mean untrusted share of the trigger positions survives augmentation at 0.516 against a typical 0.318, and on one seed all sixteen sit within the bottom 30 positions of 1024, but on another they spread to rank 1005: the ranking is intact on some runs and gone on others, and no threshold placed on it is reliable. That distinction matters for what a defender could do about it, so we tested it rather than asserting it. Replacing the fixed-deviation rule with the bimodality rule the dense experiments use changes nothing on either dataset, flagging no position on any of the six runs and leaving attack success at 94.31% and 99.47%. This is not a badly chosen constant.

The cost this leaves is real, and it falls on exactly one dataset. On CIFAR-10 the augmented model reaches 93.90% and cannot be defended while the unaugmented one reaches 87.58% and can, so a defender choosing this method gives up 6.3 points of accuracy. On GTSRB there is nothing to give up: augmentation *costs* accuracy here, 95.41% against 97.96%, so the configuration the defence needs is also the better classifier. The trade-off is a property of the dataset rather than of the method, and a defender can read it off before deciding.

G. The adaptive adversary at input-position level

The evidence in Section V-D for what a representation must preserve comes from one architecture, and the form of the method it uses is the narrower one. We therefore repeat the adversary against the input-position form: the same three sources, the same leakage construction, the same three criteria each given the budget the opinion’s own rule selects, on the ResNet-18 of Table VIII, five seeds per level.

The window reproduces, and it widens (Table X). Up to 30% leakage every criterion removes the attack, the two-state score less completely: it recovers 61% to 68% of the trigger and leaves attack success near 2%. From 35% it degrades, to 41%, 47% and 61% with recall falling from 45% to 20%, and with a spread across seeds that says it works on some runs and not on others. The two three-state criteria remove the attack at every level, at or below 0.12%, including at half

leakage, where the parameter-level form had failed at 39.8%. They coincide once more, to the second decimal at four levels of six and within 0.05 points at the other two. Discarding the compromised source and retraining fails at every non-zero leakage, 99.8% to 99.9%, exactly as at parameter level, because the poison the attacker placed in the audited source survives it.

Two things differ from the dense case, both in the defender’s favour. The two-state score’s failure is later and softer: at 30% it leaves 1.7% rather than 99.3%, and it degrades rather than collapses. And the three-state criteria hold through 50%, where the parameter-level form did not. A plausible reason, which we have not isolated, is that a position’s influence aggregates every parameter that reads it, so the untrusted share of a trigger position is a steadier quantity than that of any single weight. What we can say without qualification is that the finding of Section V-D is not a property of the fully connected model or of the parameter-level form: on a competitive architecture, with a different unit of attribution and a different intervention, two states fail and three suffice.

H. What the convolutional case shows

The extension works on four datasets and a competitive architecture, and it removes the backdoor rather than reducing it. It also widens the method’s reach: GTSRB fails the placement condition that governs the parameter-level form and is still defended here, so that condition belongs to the construction rather than to the idea. In exchange the convolutional form takes on a condition of its own, that the corruption sit at fixed input coordinates *during training* and not only in the deployed input. That condition is invisible to the dense case, it is readable from the training pipeline before any model is built, and where it holds the method is exact.

It also carries the paper’s representational claim intact: Section VI-G finds the same window as the dense experiment, wider, on this architecture. The rest of the convolutional story is the same pattern the dense results show, and Section VI-E is a clean instance of it: the attribution is a ranking, and the ranking is either right or absent. There is no regime we found where trigger positions sit in the middle of the distribution. Every failure we report is therefore a failure to *see* the trigger positions, whether because a filter excluded them or because augmentation spread their influence, and none is a failure to rank them once seen.

VII. LIMITATIONS

The corruption must be localised in input space: The defence removes parameters, so it can only help when the corruption concentrates in a small set of them. A patch trigger does. Diffuse label noise does not: when the untrusted source simply contains wrong labels, its influence is spread across every parameter, no feature stands out, and the bimodality test flags nothing (Table IV). That is the correct behaviour, but a defender should understand that a null result here means “nothing localised was found”, not “the data is clean”. Forcing removal anyway, by replacing the threshold with a fixed

budget, destroys accuracy: at a 20% budget on a label-noise model, clean accuracy falls from 96.06% to 62.96% while the attack that does not exist is unaffected. The threshold, not a budget, is the operating rule.

Large triggers cannot be removed cheaply: A 10×10 trigger spans a hundred of 784 features and overlaps regions the classifier genuinely uses. Removing enough to eliminate the backdoor costs 12 points of clean accuracy (Table III). There is a trigger size beyond which this defence is not worth applying, and where it sits depends on how much of the input the model needs.

The decision rule fires on clean models: The bimodality test selects around twenty of 784 features whether or not a backdoor is present (Table IV). We report this as a limitation even though its measured cost is at most 0.13 points of clean accuracy, because a defender reading a non-empty flag set as evidence of an attack would be wrong. The flag set is a set of features the untrusted source shaped disproportionately; on a backdoored model that is the trigger, on a clean model it is a handful of low-influence features. What carries the signal is the separation, an AUROC of 1.000 against 0.500 for the no-provenance control, not the fact that the rule fired. Turning this into a calibrated detector, with a stated false-alarm rate rather than a cost we can only report empirically, is work we have not done.

Provenance must be informative: If the defender’s opinion is vacuous about every source, there is no evidence to attribute and every parameter comes back vacuous. If it is trusted about every source, the same. The defence consumes provenance and returns nothing when given none, which Table IV shows as the chance-level control.

The trigger must sit where clean data does not: Section V-C measures this. On MNIST and Fashion-MNIST the trigger parameters take 1.000 and 0.970 of their influence from the untrusted source against typical values of 0.519 and 0.372, and the defence removes the backdoor. On GTSRB, where no region of the frame is quiet, the same quantity is 0.359 against 0.356, the rule correctly flags nothing, and the backdoor remains at 77.7%. This is adversarially exploitable: placing the trigger over active input regions removes the signal a parameter-level attribution depends on, and the condition is checkable in advance only in the sense that a defender can compute per-feature deviation on trusted data and learn whether this form of the defence is applicable at all.

The condition is not a property of provenance-based attribution as such. Attributing to input positions rather than parameters recovers the same GTSRB backdoor (Section VI-D), which means the placement condition is narrower than we first read it to be. What we cannot say is why, because that pipeline also changes the architecture and the intervention, so we report the result and not a mechanism.

Augmentation defeats the convolutional form on CIFAR-10: Section VI attributes to input positions rather than parameters, which on a ResNet-18 removes the backdoor on MNIST, Fashion-MNIST and CIFAR-10 under standard training. Random-crop augmentation translates the trigger during

training, and a position-indexed attribution assumes fixed coordinates. MNIST is unaffected and Fashion-MNIST holds on two seeds of three, but on CIFAR-10 the untrusted share of the trigger positions inverts, 0.219 against a typical 0.256, and the defence fails while the backdoor stays at 99.6%. The augmented model is also the accurate one, 93.90% against 87.58%, so on CIFAR-10 this method is bought with 6.3 points of accuracy. We have not tested architectures beyond ResNet-18, inputs larger than 32×32 , or triggers that are themselves randomised in position.

The adaptive-adversary evidence is single-dataset: The measurement of what a representation must preserve (Section V-D) is made on MNIST, on the parameter-level form, against one adversary. The argument for it is structural rather than empirical: Proposition 1 shows the collapse of all criteria is a property of single-source evidence and not of any dataset, and the leakage sweep locates the window in which two-state and three-state scores diverge. But the width of that window, and the contamination at which it opens and closes, are quantities we have measured on one dataset. Section VI-G repeats the adversary at input-position level on a ResNet-18 and finds the same window, wider, so the finding holds across two forms of the method and two architectures; it remains MNIST-only, and a second dataset could move the window’s edges.

No formal guarantee: The results are empirical. We show that a specific and widely used attack is removed under a specific and realistic knowledge assumption, and we report the conditions under which it is not.

VIII. RELATED WORK

Backdoor attacks and defences: Backdoor attacks implant trigger-conditioned behaviour through training-data poisoning [7], [8], with later work reducing the trigger’s visibility [11] or removing the need to change labels [12]. Defences differ mainly in what they assume the defender can see. Spectral signatures [1] and activation clustering [2] separate poisoned samples in a learned representation, which requires the corruption to be separable there. We evaluate both. At a 6.5% poisoning rate activation clustering removes the backdoor completely and outperforms us; at 0.97% it removes nothing, because the poisoned samples no longer form a cluster its criterion accepts (Table II). That boundary, rather than an aggregate win, is what separates the approaches. Fine-pruning [3] removes units dormant on clean data and therefore needs a clean reference set; in our setting the trigger weights feed units that are active on clean inputs, and it does not reach them (Table III). Neural Cleanse [13] recovers the trigger by optimisation, which assumes a trigger of the form it searches for. We assume none of these things and instead consume provenance, which is information a defender assembling a corpus usually has and current defences do not use.

Influence and attribution: Influence functions [4] and TracIn [5] attribute a model’s behaviour to individual training samples. Our attribution uses the same gradient decomposition, but in the opposite direction: rather than scoring samples,

we push the opinions held about samples onto parameters. Table II shows this is not a cosmetic difference. Sample-level self-influence catches under 4% of the poison here, because self-influence is largest for atypical samples while poisoned samples are deliberately easy; the same gradient information aggregated at parameter level localises the trigger exactly. We also show that with a single untrusted source our criterion reduces exactly to a scalar influence ratio, and identify the multi-source case as the one where it does not (Section V-D).

Unlearning: Removing a subset’s effect from a trained model is the goal of machine unlearning [14], typically for privacy and typically with retraining or a training-time partition prepared in advance. Our setting is the security counterpart without either: the model is given as it is, and the defender wants the effect of a source neutralised rather than certified absent.

Subjective logic in machine learning: Subjective logic [6], [15] has been used to express predictive uncertainty, most prominently in evidential deep learning [16], which places a Dirichlet over the output simplex, and in DeepTrust [17], which propagates dataset-level trust to a global model score. Both concern the trustworthiness of *outputs*. We use the formalism for a different object, the parameters, and for a specific job: keeping unvetted and compromised data on separate axes, which is exactly what an adaptive attacker exploits a two-state score for not doing (Table VI). We are careful not to claim more. A scalar built from the same three states does as well on that adversary, and what the opinion adds, a record of how much evidence a parameter carries, is not something any experiment here turns on.

IX. CONCLUSION

A defender who assembles training data from several sources knows something coarse about provenance and almost nothing about what is wrong inside any source. We showed that the coarse knowledge is enough. Attributing each parameter’s opinion to the samples that formed it, weighted by their share of the gradient, locates the parameters an untrusted contribution shaped; removing them removes an implanted backdoor without retraining, without a clean reference set, and without discarding the untrusted source’s benign data. On poisoned MNIST this drives attack success from 99.08% to 0.10% for 0.15 points of clean accuracy, where influence-ranked sample removal, the expected alternative, catches less of the poison than removing samples at random.

We do not claim the strongest defence against a conspicuous attack; activation clustering is better there and needs no provenance. The claim is about robustness of the assumption. When the attacker poisons 0.97% of the corpus and still succeeds 95.7% of the time, the representation-space defences stop finding anything, while provenance remains exactly as informative as before and the backdoor is still removed.

We were deliberate about what the representation of provenance is doing. With one untrusted source every criterion coincides, which we prove, and a defender should use the simplest. Against an adaptive attacker who also contaminates

the source the defender believes audited, what matters is that the representation keep unvetted and compromised data distinct: a score that collapses them fails completely between 30% and 40% contamination, while the opinion and a three-state scalar remove the attack there with identical numbers, at parameter level on a fully connected network and at input-position level on a ResNet-18, where the window is wider still. The subjective-logic machinery makes the three states explicit and records evidence mass; it is not what the result depends on, and we have reported it that way.

The result is conditional and we have tried to be exact about the condition. The defence acts on corruption that concentrates where clean data rarely varies. That holds on MNIST’s and Fashion-MNIST’s blank border and fails on GTSRB, whose frames carry content everywhere; it declines to act when nothing is localised, and becomes expensive when a trigger spans much of the input. Since an attacker chooses where to place a trigger, making the attribution robust to that choice is the first thing we would want to solve. Weight sharing forces a second question, because a filter is attached to no image location, and Section VI answers it by moving the unit of attribution to input positions. That move also loosens the first condition: on a ResNet-18 it removes the backdoor on all four datasets, GTSRB included, so the placement condition belongs to the parameter-level construction rather than to provenance-based attribution itself. It brings a condition of its own, that the trigger sit at fixed coordinates during training and not only at inference, which is visible in the training pipeline before a model exists. We see the broader direction as worth pursuing: provenance is information practitioners already have, records already carry, and defences currently throw away.

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We release everything needed to reproduce every number in this paper. The artifact is a single repository containing the trust-attribution implementation (Section III), the convolutional input-position variant (Section VI), the six baselines of Section IV, the driver scripts for each experiment, one per table, and the script that draws Figure 1 from the same runs, so the figure and the tables cannot disagree. Datasets are MNIST, Fashion-MNIST, GTSRB and CIFAR-10, all public and downloaded by the code on first use, and the poisoning is applied in-repository so no derived dataset needs distributing.

Each script writes a JSON summary keyed by its configuration, and the archives those scripts produced for the tables here are included alongside the code, so a reader can check our reported means and deviations without rerunning anything. The dense experiments run on a single GPU in minutes; the convolutional experiments are ResNet-18 training runs, three seeds per configuration, and take a few hours in total on one A100.

An anonymised copy of the repository will be available within three days of the submission deadline, in line with the artifact policy, and we will deposit the final version on Zenodo if the paper is accepted. We are not withholding any component.

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